

Assignment 1 — Reconstructing the Past: Estimating Solar Radio Flux F10.7 from Sunspot Records

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1. Introduction

The aim of this assignment is to study the relationship between two indicators of solar activity: the sunspot number and the solar radio flux F10.7.

The sunspot record is available for a much longer period than the F10.7 record. Therefore, the period where both measurements are available is used to study their relationship and build a model. The fitted model is then applied to earlier sunspot observations to reconstruct F10.7 for the period where direct radio-flux measurements are not available.

The datasets contain monthly mean values of sunspot number and F10.7. For further analysis, the year and month were converted into a continuous time scale, which allowed both datasets to be compared using the same time axis.

Notebook reference: cells [1]–[6] contain the introduction, data loading, separation of the columns, and conversion of year/month to decimal time.

2. Initial Monthly Data

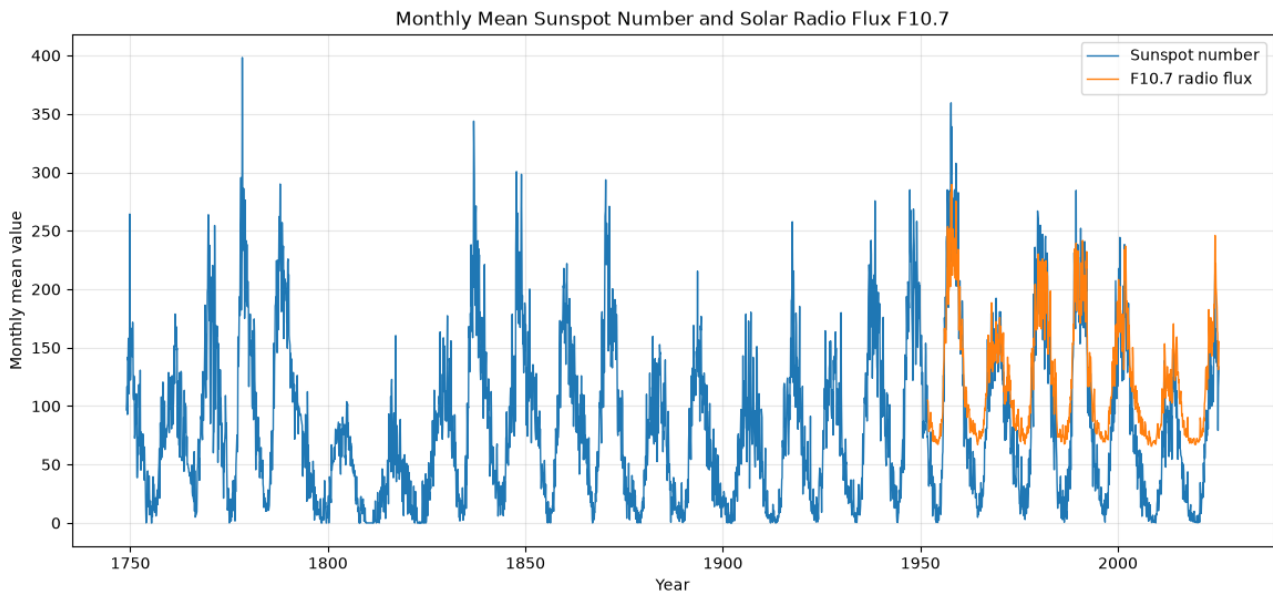
First, the original monthly mean sunspot number and F10.7 values were plotted as functions of time.

The sunspot record starts much earlier than the F10.7 record. Both indicators show clear cyclic behavior. During the period where both datasets are available, their increases and decreases generally occur at similar times.

This suggests that there is a relationship between the two indicators that can be studied further.

Notebook reference: cell [7] contains the initial monthly-data plot.

Figure 1. Monthly mean sunspot number and F10.7 radio flux.



3. 13-Month Running Mean

The monthly data contain short-term fluctuations that make the long-term behavior more difficult to compare. Therefore, a 13-month running mean was applied to both datasets.

13-month running mean $\bar{R}$

$$\bar{R} = \frac{1}{24}R_{i-6} + \frac{1}{12}(R_{i-5} + R_{i-4} + \cdots + R_{i-1} + R_i + R_{i+1} + \cdots + R_{i+5}) + \frac{1}{24}R_{i+6}$$

For each central month, the calculation uses six months before the current month, the current month, and six months after it. The two values at the edges have half weight.

For the first and last six months of the data, only the available neighboring values are averaged.

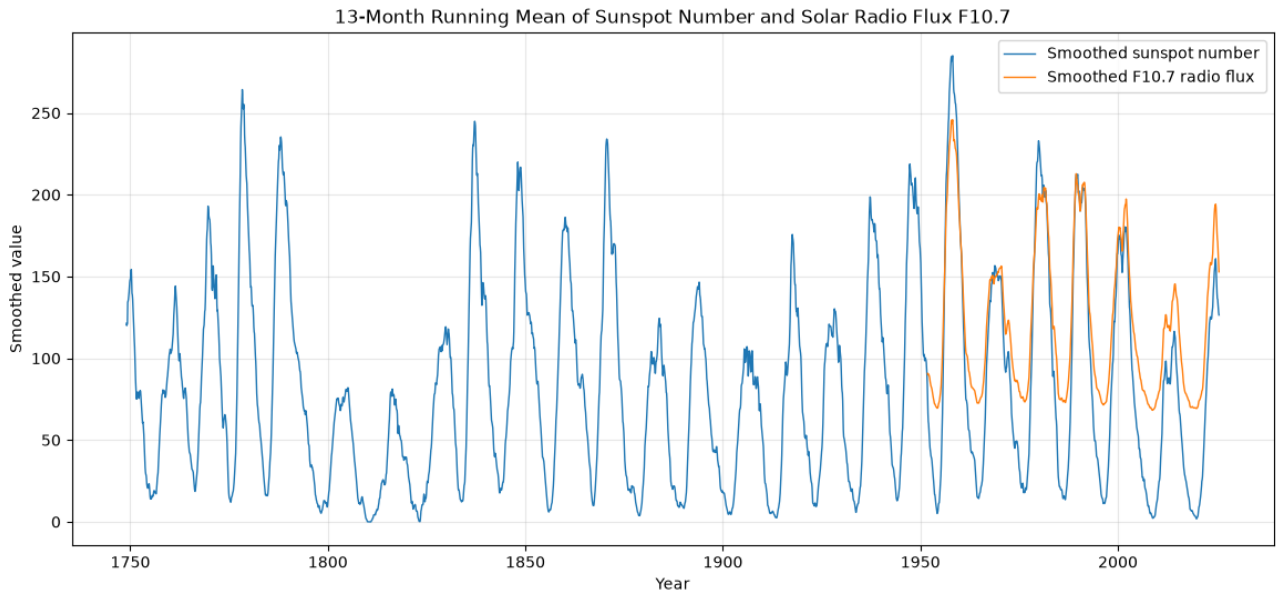
The smoothing procedure was implemented manually, without using a built-in moving-average function.

Notebook reference:

- cell [8] — implementation of the running mean;
- cell [9] — application of smoothing to both datasets.

After smoothing, short-term fluctuations are reduced while the main cyclic behavior remains visible. This makes the long-term relationship between sunspot number and F10.7 easier to see.

Figure 2. Sunspot number and F10.7 after the 13-month running mean.



4. Common Observation Period and Relationship Between the Variables

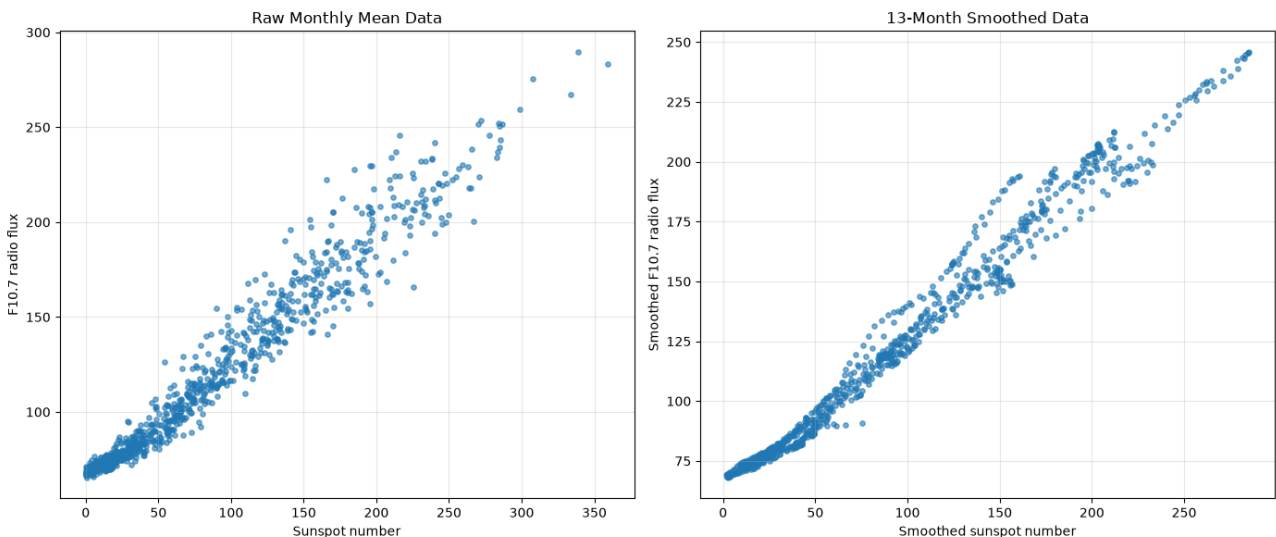
The two datasets cover different time intervals. Therefore, before fitting the model, the common period where both measurements are available was identified.

The common observation period starts in November 1951 and ends in August 2025.

Both datasets contain 886 monthly observations during this period.

The corresponding time points were also checked to make sure that each sunspot value is compared with the F10.7 value from the same month.

Figure 3. Raw and smoothed relationship between sunspot number and F10.7.



Notebook reference:

- cell [11] — calculation of the common period;
- cells [12-14] — selection and checking of the common data.

The time-point check confirmed that the two datasets are correctly aligned.

To examine the relationship between sunspot number and F10.7, the original and smoothed common-period data were compared using scatter plots.

Each point represents one month. The horizontal coordinate is the sunspot number and the vertical coordinate is F10.7 for the same month.

The raw monthly data show a positive relationship but contain more scatter. After smoothing, the points are closer to the main trend.

This means that larger sunspot numbers are generally associated with larger F10.7 values.

5. Regression Model

The smoothed data from the common observation period were used to construct a third-order regression model.

The sunspot number is used as the input variable and the smoothed F10.7 value is used as the output variable.

The squared and cubed sunspot-number terms allow the model to describe a curved relationship between the two indicators.

Multi-dimensional linear regression model:

$$F_i = \beta_0 + \beta_1 R_i + \beta_2 R_i^2 + \beta_3 R_i^3 + \varepsilon_i \quad (1)$$

Where,

- F is the vector of dependent variables (regressand), representing the smoothed solar radio flux at 10.7 cm.

$$F = \begin{bmatrix} f_1 \\ f_2 \\ \dots \\ f_N \end{bmatrix}$$

and $f_1, f_2, \dots, f_N$ is the smoothed flux at different times.

- R is the matrix of independent variables (regressors)

$$R = \begin{bmatrix} 1 & r_1 & r_1^2 & r_1^3 \\ 1 & r_2 & r_2^2 & r_2^3 \\ \dots & \dots & \dots & \dots \\ 1 & r_N & r_N^2 & r_N^3 \end{bmatrix}$$

and $r_1, r_2, \dots, r_N$ are the smoothed sunspot numbers at different times.

- β is the vector of coefficients to be estimated

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix}$$

- ε_i is the error term

Notebook reference:

- cell [17-20] — calculation of the squared and cubed terms.

For every monthly observation, the regression matrix contains four terms: the constant term, the sunspot number, its square, and its cube.

The resulting regression matrix contains 886 rows and 4 columns.

6. Least Squares Estimation

The coefficients of the regression model were calculated using the Least Squares Method.

$$\beta = (R^T R)^{-1} R^T F$$

In the calculation, the regression matrix was first transposed.

Notebook reference:

- cell [21] — transpose, inverse;
- cell [22] — final calculation of the regression coefficients.

The estimated coefficients were:

$$\beta_0 \approx 65.7627$$

$$\beta_1 \approx 0.4674$$

$$\beta_2 \approx 0.0019706$$

$$\beta_3 \approx -5.2233 \times 10^{-6}$$

These coefficients define the fitted relationship between the smoothed sunspot number and F10.7.

7. Reconstruction on the Measured Period

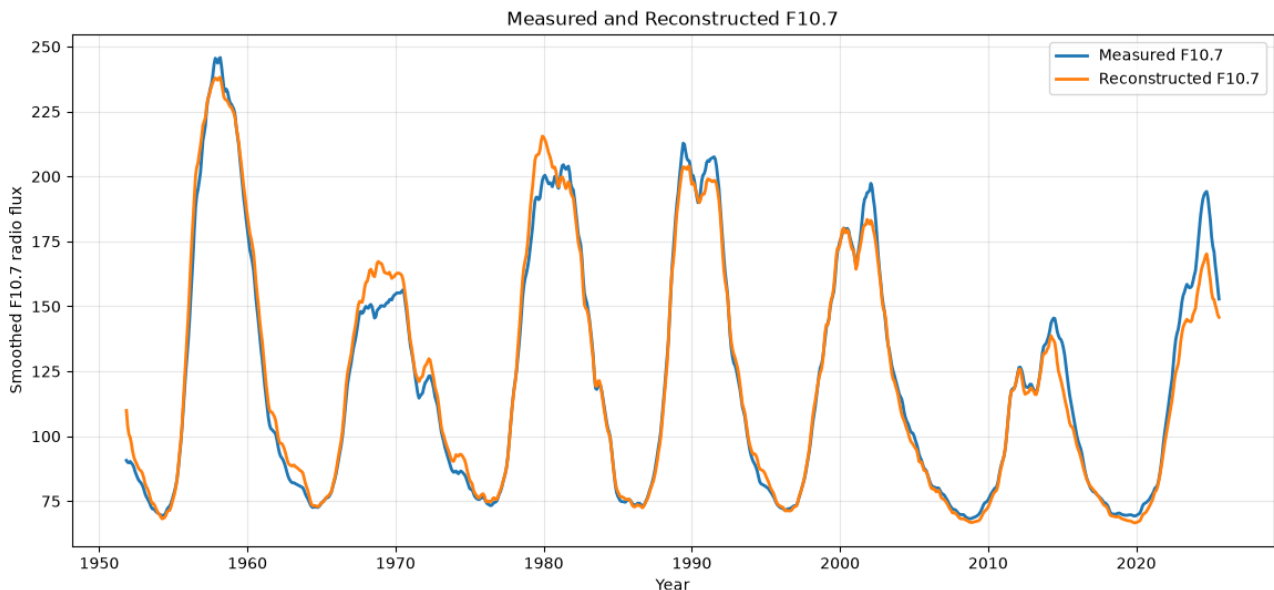
The fitted regression model was first applied to the common observation period, where both sunspot number and real F10.7 measurements are available.

This produced one reconstructed F10.7 value for every monthly sunspot observation.

$$F_i = \beta_0 + \beta_1 R_i + \beta_2 R_i^2 + \beta_3 R_i^3$$

This is the regression equation without the error term, used to calculate reconstructed F10.7.

Figure 4. Measured and reconstructed F10.7.



Notebook reference:

- cell [23] — calculation of reconstructed F10.7;
- cell [24] — comparison between measured and reconstructed F10.7.

The reconstructed curve follows the main increases, decreases, peaks, and minima of the measured curve.

However, the curves are not identical. Therefore, the difference between measured and reconstructed values was evaluated numerically.

8. Reconstruction Error

For each month, the difference between the measured and reconstructed F10.7 values was calculated.

A positive error means that the model underestimated the measured value, while a negative error means that the model overestimated it.

The overall error was evaluated using RMSE.

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (f_i - \hat{f}_i)^2}$$

Notebook reference:

- cell [25] — calculation of the monthly errors;
- cell [26] — complete RMSE calculation.

The result was:

$N = 886$

$RMSE \approx 6.29$

RMSE provides one numerical value describing the scale of the difference between measured and reconstructed F10.7 over the common observation period.

This error value was also used to estimate the uncertainty of the historical reconstruction.

9. Historical Reconstruction of F10.7

The fitted model was then applied to the earlier sunspot record, where direct F10.7 measurements are not available.

The historical reconstruction covers approximately the period from 1749 to October 1951.

The historical period contains 2434 monthly sunspot observations.

Notebook reference: cell [27] contains the selection of the historical period and calculation of the reconstructed historical F10.7 values.

The same regression equation and coefficients obtained from the common observation period were used for the historical reconstruction.

$$F_i = \beta_0 + \beta_1 R_i + \beta_2 R_i^2 + \beta_3 R_i^3$$

10. Uncertainty of the Historical Reconstruction

The reconstructed historical values are estimates rather than direct measurements.

To show the uncertainty of the reconstruction, an interval of ± 3 RMSE was used.

The calculated value was approximately:

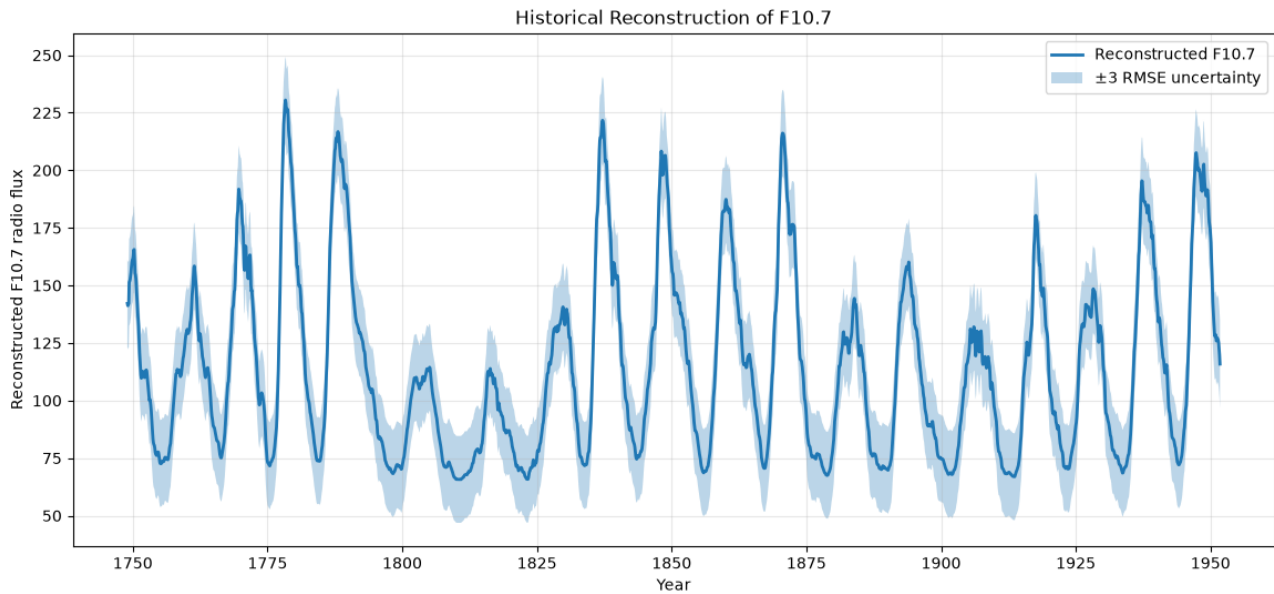
$3 \times RMSE \approx 18.86$

The lower and upper limits were obtained by subtracting and adding this value to the reconstructed F10.7.

Notebook reference:

- cell [28] — calculation of lower and upper uncertainty bounds;
- cell [29] — historical reconstruction plot.

Figure 5. Historical reconstruction of F10.7 with ± 3 RMSE uncertainty.



The reconstructed historical series follows the long-term cyclic behavior of the sunspot record.

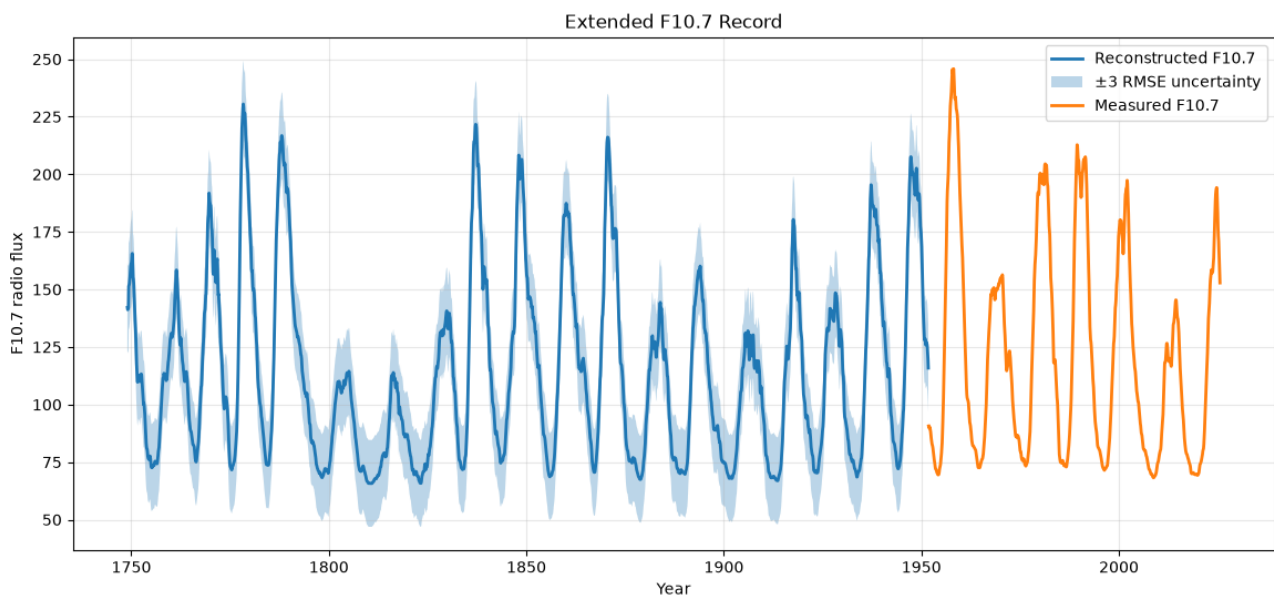
The shaded area represents the uncertainty range around the reconstructed values.

The historical F10.7 values should therefore be interpreted as estimates rather than direct measurements.

11. Extended F10.7 Record

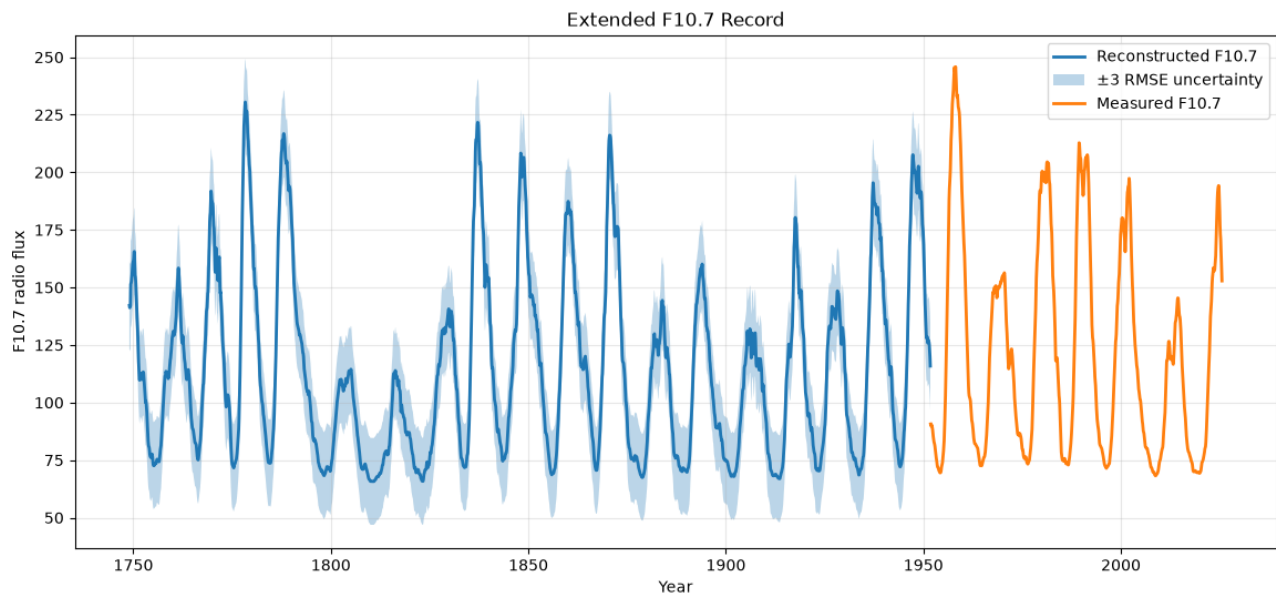
Finally, the historical reconstruction and the directly measured F10.7 record were shown on the same time axis.

Figure 6. Extended F10.7 record.



Notebook reference: cell [30] — extended F10.7 record.

The blue curve shows the reconstructed historical F10.7 values, while the orange curve shows the directly measured F10.7 values. The shaded area represents the ± 3 RMSE uncertainty range of the historical reconstruction.



The combined plot shows the reconstructed historical period followed by the period where direct F10.7 measurements are available.

12. Conclusions

The assignment covered the main steps of experimental data processing: preparing the data, smoothing, building a model, estimating the error, and reconstructing missing values.

The 13-month running mean reduced short-term fluctuations and made the relationship between sunspot number and F10.7 easier to see.

Using the common observation period, a third-order regression model was constructed using the Least Squares Method.

The reconstructed F10.7 followed the measured data reasonably well. The RMSE was approximately 6.29.

The model was then used to reconstruct F10.7 for the earlier period, when only sunspot data were available. The uncertainty range was approximately ± 18.86 .

Overall, the assignment showed how one measured quantity can be used to estimate another quantity for a period where direct measurements are unavailable.

13. Learning Log

In this assignment, we worked with data preprocessing, smoothing, regression, error estimation, and historical reconstruction of F10.7.

The main difficulties were correctly aligning the two datasets, understanding the 13-month smoothing procedure, and interpreting the regression and RMSE results.

We addressed these issues by checking the common observation period, comparing raw and smoothed data, reviewing the regression calculation step by step, and comparing reconstructed values with measured F10.7.

As a result, we learned how to process experimental data, build a model, estimate its error, and reconstruct values for a period where direct measurements are not available.

14. Team Contribution

The assignment was completed by a team of three students.

The work was divided between data preparation and visualization, implementation of the main calculations, and analysis of the obtained results.

Each team member worked on a separate part of the assignment, while the main intermediate results were discussed and checked together.

The final reconstruction, error estimation, conclusions, and report were reviewed jointly before submission.